

nopol' and American 'Copperhead' are well-known examples, and the former is in service with the Indian Army.

GPS-guided shells

Typically, these shells have a maximum range of 35 kilometres. The shell's guidance system is centred on a mission computer that continuously derives the GPS coordinates from a shell-embedded GPS receiver. As in the laser-guided shells, fins are present in this shell also. Before firing a shell, the target's GPS coordinates are loaded into the shell's mission computer by a trooper known as 'fuse setter.' When the shell is fired, the embedded GPS receiver constantly feeds current GPS coordinates to the mission computer, which compares them with the pre-loaded coordinates. It then calculates the required trajectory and guides the shell in that trajectory.

The advantage of this is that, even if the shell is shot off-target, it automatically glides toward the target. When fired in MRSI mode, these shells can raze a single building to the ground without any damage to neighbouring buildings. The American 'Excalibur (M982),' a product of Raytheon Missile Systems, and 'Bofors' of BAE Systems, are the only known examples. Typically, the estimated cost of a single shell is around US\$ 50,000!

Sensor-fused tank killer shells

Meant to take out enemy tanks, these shells act as carriers and deploy smaller, standalone, smart anti-tank munitions that take out tanks. Each smart submunition contains a flight computer, radar and an explosive detonating mechanism. When the shell reaches the target area, submunitions are ejected out and slowly descend through a parachute and spin. While spinning, the submunition's radar is activated. The radar maintains a spiral-scanning pattern during descent. When the flight computer recognises the radar echoes of a tank, the computer simply jettisons the parachute. This allows the submunition to fall over the target. An explosive detonating mechanism detonates the explosive. The result is an explosively-formed penetrator (EFP), a molten-copper jet travelling at hyper-



Fig. 16: Russian krasnapol laser-guided artillery shell (Photograph courtesy: 'Mike1979 Russia' through Wikipedia)



Fig. 17: Excalibur GPS guided round (Photograph courtesy: US Army, through Wikipedia)

sonic speeds. This jet penetrates the tank's weakest place—its top. In case the submunition is not able to find any tank, it automatically explodes and destructs itself. The American 'SADARM M898,' German 'Suchzünder Munition für die Artillerie 155' (meaning 'sensor-fused munition for 155mm artillery') and 'BONUS' of BAE systems are the presently operational shells.

An entire armoured offensive of the enemy could be blunted by batteries of howitzers firing such sensor-fused shells.

Jammer shells

These shells are of a different genre. They consist of a high-powered radio frequency (RF) jammer that jettisons through a parachute and lands in the target area. After impacting the ground, it starts emitting high-power noise in a particular RF band. This turns all communication systems that use that particular band in the target area 'deaf and dumb.' Less than a dozen of these rounds, jamming various bands, can completely choke all communication systems for a definite period. The military application of this system is vast and spreads over various doctrines. So far, such shells are said to be available with the Russian forces in limited scale.

Command and control systems

Each combat leader in the echelon gets his orders from his superior based on ground situation, delegates a task to his subordinate units and reports back the result to his superior. This process is called command and control (C²) and is

based on robust communication systems. The C² is to be maintained from the top to the bottom level forces—be it a 4-6 member team or 40,000-50,000 strong strike corps. The performance of a fighting force is directly influenced by the efficacy of C². Tanks and SPGs, crucial to a battle, are highly communication-intensive systems.

Communication systems for tanks

With improvements in communication technology, a better C² has become possible, from intra-tank communication to inter-tank communication. Due to the immense noise associated with the tanks, direct voice communication is not even possible between the tank crew seated next to each other. Helmet-mounted head-phone intercoms are used for intra-tank voice communication. These are plugged into a wired two-way communication system. With the inter-tank communication mode selected, the tank commander can communicate with other tank commanders in the formation or his superiors using a wireless radio transceiver. The warning from the active protection system (APS) is coupled with the intercom system. The commander hears a warning tone if his tank is targeted—an eventual sign of an impending missile attack. A specially-designated tank with a better communication facility, called command tank, is also used. This tank hosts the formation commander, who commands the tank commanders of the formation. He or she also communicates with the higher echelons.

Due to the changing geopolitics, combined infantry-armour operations are required inside the cities. This requires the infantry to move in close proximity with the tanks. To facilitate better infantry vs tank communication,